

Two-Point Method: A Quick and Fatigue-Free Procedure for Assessment of Muscle Mechanical Capacities and the 1 Repetition Maximum

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Supported by the Serbian Research Council under Grant 175037.

ABSTRACT

THE FORCE-VELOCITY RELATIONSHIP IS USED TO EVALUATE THE MAXIMAL CAPACITIES OF ACTIVE MUSCLES TO PRODUCE FORCE, VELOCITY, AND POWER. THE LOAD-VELOCITY RELATIONSHIP HAS ALSO BEEN USED TO PREDICT THE 1 REPETITION MAXIMUM (1RM) AND, CONSEQUENTLY, THE RELATIVE LOADS (%1RM). THE 2-POINT METHOD HAS RECENTLY EMERGED AS A QUICK AND FATIGUE-FREE PROCEDURE FOR THE ASSESSMENT OF BOTH THE FORCE-VELOCITY AND LOAD-

VELOCITY RELATIONSHIPS. THIS ARTICLE SUMMARIZES THE RESULTS OF PREVIOUS STUDIES THAT HAVE EXPLORED THE FEASIBILITY OF THE 2-POINT METHOD AND PROVIDES PRACTICAL RECOMMENDATIONS FOR ITS APPLICATION BY STRENGTH AND CONDITIONING PROFESSIONALS.

INTRODUCTION

The force-velocity (F-V) and load-velocity (L-V) relationships have received the attention of the scientific community for a long time (38,55). More recently, the proliferation of affordable sport technology has allowed strength and conditioning professionals to routinely

evaluate these relationships in various movement tasks (1,3). The F-V relationship provides a higher informational value than routinely applied tests typically conducted under a single mechanical condition because it allows to discern among different muscle mechanical capacities (23,37). However, the L-V relationship has been used to predict the 1 repetition maximum (1RM) (14,17,24,31). Therefore, the individual assessment of the F-V and L-V relationships is highly

KEY WORDS:

force-velocity relationship; load-velocity relationship; muscle function; 2-load method; 2-velocity method; velocity-based resistance training

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recommended to be used for prescribing and implementing effective individualized resistance training programs. However, both relationships have been obtained from different mechanical conditions that made the related procedures not only long lasting but also prone to fatigue. Therefore, the objective of this article is to present the 2-point method (i.e., the method based on tests performed under only 2 loads or 2 velocities) as a quick and fatigue-free procedure able to reliably and accurately determine both F-V and L-V relationships. We will discuss the theoretical basis and the most important methodological issues related to the application of the 2-point method for obtaining reliable and valid F-V and L-V relationships.

THEORETICAL BASIS

FORCE-VELOCITY RELATIONSHIP AND ITS PARAMETERS

It is generally accepted that muscles have distinctive mechanical capacities that allow them to produce maximal levels of force (F_0), velocity (V_0), and power (P_0) (27). The main limitation of the standard testing procedures conducted under a single mechanical condition (e.g., a vertical jump performed against a preselected load) is that their single outcomes do not allow for distinguishing among different mechanical capacities of the muscles (27). It should be noted that the force, velocity, and power outputs obtained under a single load are interdependent (i.e., higher force outputs inevitably produce a higher velocity and, consequently, higher power outputs) (13,27). However, an important characteristic of the F-V approach is that F_0 and V_0 are independent of each other because a higher F_0 does not necessarily imply a higher V_0 . A number of studies have revealed that the parameters of the typically linear F-V relationship (i.e., F_0 , V_0 , a , and P_0) are reliable and at least moderately valid (9,13,15,28,53). Briefly, to obtain the F-V relationship parameters, the force and velocity data collected from a functional movement (e.g., jumping, cycling, lifting, running,

pushing, etc.) conducted under 2 or more loading or velocity conditions provide a line that determines the following F-V equation: $F(V) = F_0 - aV$, in which F_0 represents the force intercept (i.e., force at zero velocity) and a is the slope of the F-V relationship (27,48). The maximum velocity (V_0 ; velocity at zero force) corresponds to F_0/a . Finally, because of the high linearity of the F-V relationship, P_0 can be calculated as $P_0 = F_0 \cdot V_0/4$. Therefore, the F-V relationship should be recommended for the routine testing of muscle function because it provides higher informational value than the standard test conducted under a single mechanical condition (27,28,48).

The F-V relationship parameters have been used to gain a deeper insight into different issues related to muscle function, rehabilitation, injury prevention, and sport performance. Specifically, it has been shown that F-V relationship parameters (i.e., F_0 , V_0 , a , and P_0) may distinguish between individuals of different physical fitness levels (10) and age (56), as well as between athletes of different disciplines (22). The F-V profile may also partly explain between-subjects differences in bilateral deficit during ballistic lower limb push-off (49). Finally, there may be a subject-specific optimum balance between F_0 and V_0 capacities (i.e., the F-V slope), which maximizes ballistic performance for a given value of P_0 (32,33,46). Therefore, the training programs should target a minimization of the F-V imbalance (37). Taken collectively, these results highlight the importance of the F-V relationship assessment not only for basic research in various fields of human movement science but also for better prescription and monitoring of resistance training programs by strength and conditioning practitioners.

LOAD-VELOCITY RELATIONSHIP AND 1 REPETITION MAXIMUM

The 1RM is one of the most common measurements used by strength and conditioning professionals either for assessing the effects of training procedures or for prescribing specific

relative loads (i.e., %1RM) (8,52). However, the direct determination of the 1RM typically involves a number of limitations, such as being prone to fatigue and injuries, being time-consuming and impractical for large groups of subjects, whereas 1RM value can change quickly as a consequence of training. To solve these problems, several indirect methods have been proposed to predict the 1RM. The prediction of the 1RM through the L-V relationship has attracted considerable attention over recent years (24,31,34,36). The L-V relationship, interpreted as the relationship between the %1RM and movement velocity, has been explored in a variety of resistance training exercises such as the bench press, bench pull, pull-up, squat, or leg press (7,17,24,39,50,51). The strong and approximately linear L-V relationship observed in the cited studies motivated a proposal for generalized group equations to predict the %1RM from the recorded velocity output (7,17,24,39,50,51). Specifically, the proposed equations enable an estimation of 1RM and %1RM from a single repetition conducted against a given load provided that it is performed at maximal velocity (23,24).

The information presented above suggests that testing the movement velocity is a safe and time-saving method of predicting the 1RM. However, generalized group equations are associated with at least 4 important drawbacks that limit their use in practice. First, an independent generalized group equation would be required for each exercise because the L-V relationship is inevitably exercise-specific (7,50). Second, although the differences in the L-V relationship between different exercises may be caused by the differences in the velocity of the 1RM, recent studies have shown that the inclusion of the stretch-shortening cycle significantly increases the velocity associated to each %1RM compared with the concentric-only execution (17,40,42). Third, a systematic bias in the velocity outputs may exist between different devices

commonly used to measure movement velocity (2). Therefore, the generalized group equations could also be dependent on the selection of measurement tools. Finally, in line with the results obtained for the F-V relationship, the L-V relationship could also be subject-dependent (17). The listed drawbacks of generalized group equations could be solved through the individual modeling of the L-V relationship (3,17,31). However, the standard tests used to determine the load-velocity relationship are based on multiple external loads (usually between 5 and 9), which makes them prone to fatigue and, therefore, impractical when undertaken on a daily basis (7,24,31). Therefore, a potential solution to overcome these limitations could be the application of the 2-point method to individually predict the 1RM.

LINEARITY OF THE F-V AND L-V RELATIONSHIPS

The most important characteristic of the F-V and L-V relationships of functional multijoint tasks (e.g., jumping, lifting, running, cycling, etc.) is that both of them are highly linear (18,27). The linearity of F-V relationship of human muscles acting within a multijoint system differs from the hyperbolic F-V relationship typically described in studies of in vitro animal muscles and individual human muscle groups (6,27). In this regard, it should be noted that previous studies have found only trivial differences in the strength of the F-V relationship between linear and polynomial regression models in multijoint exercises such as vertical jumps (9) and the bench press throw (53). It could be argued that the linearity of the F-V relationship is a consequence of a narrow range of force and velocity data used for its modeling. Unfortunately, certain functional movements impose severe limitations on the magnitudes of the loads that can be applied. For example, although in vertical jumps, the experimental points are inevitably force-biased (i.e., close to the force intercept) because of the inevitable role of body mass (9,13,33), in the leg cycle

ergometer exercise, they are velocity-biased (i.e., close to the velocity intercept) (20,54). Moreover, none of the tested movements can be performed at velocities close to their theoretical maximum because that would imply both a negligible load and “instant” muscle activation. Consequently, neither of the 2 prerequisites can be accomplished. Nevertheless, it could be safely argued that the observed experimental findings do not provide reasons for expecting that F-V relationships would prominently deviate from linearity if a wider range of loads of velocities were tested.

The strength of the individual L-V relationship has also proven to be high for a variety of resistance training exercises such as the bench press ($R^2 \approx 0.99$) (18,50), bench press throw ($R^2 \approx 0.98$) (18), bench pull ($R^2 \approx 0.99$) (50), pull-up ($R^2 \approx 0.98$) (39), squat ($R^2 \approx 0.98$) (42), and vertical jumps ($R^2 \approx 0.98$) (42). These results provide support for using the linear regression model instead of more complex ones (6). Note that the linear regression model should allow for a more time-efficient procedure because a lower number of experimental points would be needed to model the individual F-V and L-V relationships.

MULTIPLE-POINT METHOD VS. 2-POINT METHOD

The F-V relationship parameters (i.e., F_0 , V_0 , a , and P_0) obtained from the multiple-point method in different functional multijoint tasks have repeatedly proven to be reliable and at least moderately valid (28). Studies have also revealed that the F-V relationship may be sensitive enough to detect the effects of a short-term resistance training program, in which training with heavy and light loads were associated with improvements in F_0 and V_0 , respectively (11,20,32). The multiple-point method applied to the individual L-V relationship has also demonstrated an accurate estimation of the 1RM value (3,31,44). However, it should be noted that the multiple-point method may be time-consuming and, consequently, prone to fatigue. In this regard,

the 2-point method has emerged with the expectation of providing the same information but considerably reducing the testing time and the associated fatigue (28). The high linearity of F-V and L-V relationships of functional multijoint tasks may support the feasibility of the 2-point method for assessment of muscle mechanical capacities and the 1RM. Figure 1 depicts the high overlap of the F-V and L-V relationships commonly observed when obtained from the multiple-point and 2-point methods.

Because of the high linearity of both the F-V and L-V relationships, it was hypothesized that the addition of intermediate points (i.e., either loads or velocities to the 2 already tested) would not improve the precision of the relationship modeled (28,43,57). To date, only 6 studies have compared the basic properties, such as the reliability, validity, and sensibility of the 2-point method against the same properties of the multiple-point method, where more than 2 points are used for the regression modeling (Table 1) (12,20,25,41,43,57). Collectively, these results revealed trivial differences both in the magnitude and the reliability of the F-V relationship parameters between the 2 methods. In addition, the F-V relationship parameters obtained from the 2-point method have shown exceptionally strong relationships with the same parameters obtained from the multiple-point method. García-Ramos et al. (20) recently demonstrated that the 2-point method could be used to detect the selective changes on the F-V relationship parameters induced by a cycle ergometer sprint training program. These results strongly encourage the use of the 2-point method to determine the F-V relationship because it provides virtually identical outcomes as the multiple-point method and with comparable reliability. Finally, although the comparison of the accuracy of 1RM prediction between the multiple-point and 2-point methods has not been explored yet, it is plausible to assume that due to a typically high linearity of

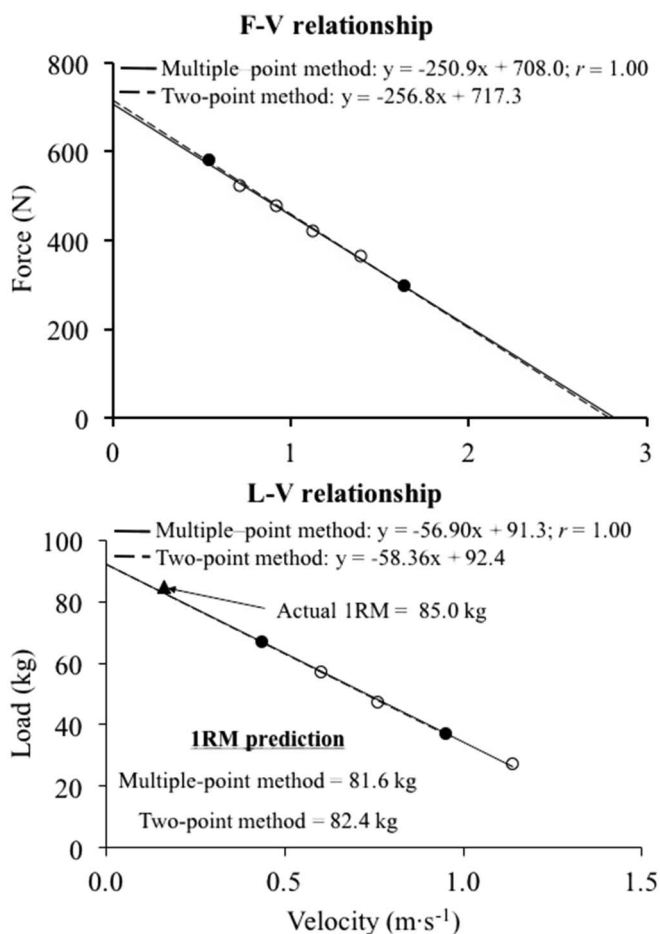


Figure 1. Force-velocity (F-V; upper panel) and load-velocity (L-V; lower panel) relationships depicting the high overlap existing when obtained from the multiple-point (straight lines) and 2-point (dashed lines and filled dots) methods. The F-V relationships (adapted with permission from Pérez-Castilla et al. (43)) were obtained from the bench press throw exercise from the data recorded from 6 different loads (multiple-point method: 20% 1RM, 30% 1RM, 40% 1RM, 50% 1RM, 60% 1RM, and 70% 1RM) and averaged across the subjects ($n = 22$) and from only the 2 extreme loads (2-point method: 20% 1RM and 70% 1RM). The L-V relationships of an individual subject were determined during an incremental loading test in the bench press exercise from 5 different loads (multiple-point method: 32% 1RM, 44% 1RM, 55% 1RM, 67% 1RM, and 79% 1RM) and only 2 loads (2-point method: 44% 1RM and 79% 1RM). The 1RM was estimated from the 2 equations depicted as the load associated with a mean velocity of $0.17 \text{ m} \cdot \text{s}^{-1}$. The directly measured 1RM (filled triangle) is also depicted but is not included in the multiple-point method regression. r = Pearson's correlation coefficient.

the L-V relationship, both methods should produce almost identical outcomes.

The outcomes related to the 2-point method presented in Table 1 have been obtained from the 2 farthest data points (i.e., the highest and the lowest loads or velocities) that were used for the

modeling of the multiple-point method. The use of the 2 farthest data points seems appropriate because previous studies have shown that the reliability and validity of the F-V relationship may be compromised when the distance between the 2 experimental points used for the F-V

modeling is reduced (43). Within the next section, an overview of the methodological elements that need to be considered when applying the 2-point method in routine testing is provided.

HOW TO APPLY THE 2-POINT METHOD IN ROUTINE TESTING

OPTIMAL SELECTION OF THE 2 POINTS

Distance between experimental points. Only 1 investigation has specifically assessed the effect of the distance between experimental points on the reliability and validity of the 2-point method (43). Specifically, 3 different 2-point methods were constructed in the bench press throw exercise: 20–70% 1RM (i.e., farthest data points), 30–60% 1RM, and 40–50% 1RM (i.e., closest data points). The main outcome of this investigation was a decrease in both the reliability and validity of the F-V relationship parameters associated with the proximity of the data points: 40–50% 1RM (coefficient of variation [CV] = 18.0%; $r = 0.64$) < 30–60% 1RM (CV = 7.3%; $r = 0.94$) < 20–70% 1RM (CV = 5.5%; $r = 0.98$). The practical recommendation based on these data would be that the routine application of the 2-point method should be based on the selection of the 2 most distant loads or velocities.

Proximity to the force and velocity intercepts. Pérez-Castilla et al. (43) evaluated the effect of the location of the 2 loads with respect to the force and velocity axis on the reliability and validity of the F-V parameters. Although these results were less conclusive, they suggested that although force-biased data points (50–70% 1RM) may provide F_0 with higher accuracy, velocity-biased data points (20–40% 1RM) could be preferable to gain precision in the determination of V_0 . The practical recommendation would be, if possible, to select 1 point close to V_0 and another 1 close to F_0 .

Reliability of individual points. The individual points with low reliability should be avoided because they may

Table 1
Comparison of the F-V relationship parameters obtained from the multiple-point and the 2-point methods

Study	Protocol	F-V parameter	Method	Magnitude	Reliability	Validity
Pérez-Castilla et al. (41)	23 collegiate men performed 2 tests in the CMJ exercise against 6 loads: 0, 17, 30, 45, 60, and 75 kg.	F_0 (N)	Multiple	3218 ± 356	CV: 2.7%; ICC: 0.95	ES: -0.02 ; r : 0.99
			2-point	3226 ± 415	CV: 3.0%; ICC: 0.95	
		V_0 ($\text{m} \cdot \text{s}^{-1}$)	Multiple	5.69 ± 0.92	CV: 5.8%; ICC: 0.89	ES: -0.21 ; r : 0.96
			2-point	5.89 ± 0.97	CV: 7.2%; ICC: 0.81	
		a ($\text{N} \cdot \text{s} \cdot \text{m}^{-1}$)	Multiple	583 ± 125	CV: 10.1%; ICC: 0.88	ES: 0.11; r : 0.95
			2-point	568 ± 138	CV: 11.9%; ICC: 0.82	
		P_0 (W)	Multiple	4531 ± 523	CV: 3.0%; ICC: 0.95	ES: -0.29 ; r : 0.98
			2-point	4679 ± 509	CV: 4.3%; ICC: 0.88	
Pérez-Castilla et al. (43)	21 collegiate men performed 2 tests in the bench press throw exercise against 6 loads: 20% 1RM, 30% 1RM, 40% 1RM, 50% 1RM, 60% 1RM, and 70% 1RM.	F_0 (N)	Multiple	708 ± 108	CV: 4.15%; ICC: 0.94	ES: -0.11 ; r : 0.99
			2-point	720 ± 112	CV: 4.45%; ICC: 0.94	
		V_0 ($\text{m} \cdot \text{s}^{-1}$)	Multiple	2.86 ± 0.39	CV: 6.04%; ICC: 0.72	ES: 0.08; r : 0.97
			2-point	2.83 ± 0.37	CV: 6.28%; ICC: 0.67	
		a ($\text{N} \cdot \text{s} \cdot \text{m}^{-1}$)	Multiple	252 ± 52	CV: 8.50%; ICC: 0.84	ES: -0.13 ; r : 0.98
			2-point	259 ± 53	CV: 9.02%; ICC: 0.83	
		P_0 (W)	Multiple	2026 ± 385	CV: 4.34%; ICC: 0.95	ES: -0.02 ; r : 0.98
			2-point	2034 ± 390	CV: 4.67%; ICC: 0.94	

**Table 1
(continued)**

García-Ramos et al. (19)	27 collegiate men performed 2 tests in the leg cycle ergometer exercise against 4 resistances: 0.40, 0.63, 0.86, and 1.09 N·kg ⁻¹ of body mass.	F ₀ (N·kg ⁻¹)	Multiple	13.7 ± 1.3	CV: 3.57%; ICC: 0.87	ES: 0.05; r: 0.99
			2-point	13.6 ± 1.2	CV: 3.41%; ICC: 0.88	
		V ₀ (m·s ⁻¹)	Multiple	3.78 ± 0.13	CV: 1.47%; ICC: 0.89	ES: 0.20; r: 0.96
			2-point	3.75 ± 0.14	CV: 1.46%; ICC: 0.89	
		a (N·s·m ⁻¹ ·kg ⁻¹)	Multiple	3.62 ± 0.36	CV: 4.40%; ICC: 0.82	ES: -0.03; r: 0.98
			2-point	3.63 ± 0.35	CV: 4.09%; ICC: 0.85	
		P ₀ (W·kg ⁻¹)	Multiple	12.9 ± 1.3	CV: 3.09%; ICC: 0.91	ES: 0.10; r: 0.98
			2-point	12.8 ± 1.3	CV: 3.25%; ICC: 0.91	
Zivkovic et al. (57)	12 healthy men performed a test in the CMJ exercise against 0, 8, 16, 24, and 32 kg.	F ₀ (N)	Multiple	2948 ± 788	No data	ES: -0.09; r: 0.99
			2-point	3020 ± 896		
		V ₀ (m·s ⁻¹)	Multiple	4.5 ± 1.8		ES: 0.00; r: 0.99
			2-point	4.5 ± 1.7		
		P ₀ (W)	Multiple	3089 ± 584		ES: -0.04; r: 0.98
			2-point	3109 ± 555		
	12 healthy men performed a test in the leg cycle ergometer exercise against 2, 4, 6, 8, and 10 kg.	F ₀ (N)	Multiple	1033 ± 222	No data	ES: 0.00; r: 0.99
			2-point	1032 ± 223		
		V ₀ (m·s ⁻¹)	Multiple	4.1 ± 0.3		ES: 0.33; r: 0.99
			2-point	4.0 ± 0.3		
		P ₀ (W)	Multiple	1047 ± 189		ES: 0.09; r: 0.99
			2-point	1030 ± 188		

(continued)

Table 1
(continued)

	12 healthy men performed a test in the bench press throw exercise against 20, 27.5, 35, 42.5, 50, and 57.5 kg.	F_0 (N)	Multiple	830 ± 58	No data	ES: -0.42 ; r : 0.95
			2-point	856 ± 66		
		V_0 ($\text{m} \cdot \text{s}^{-1}$)	Multiple	3.0 ± 0.3		ES: 0.00; r : 0.95
			2-point	3.0 ± 0.3		
		P_0 (W)	Multiple	627 ± 73		ES: -0.20 ; r : 0.98
			2-point	642 ± 77		
	12 healthy men performed a test in the bench press pull exercise against 20, 27.5, 35, 42.5, 50, and 57.5 kg.	F_0 (N)	Multiple	1350 ± 234	No data	ES: -0.06 ; r : 0.99
			2-point	1364 ± 246		
		V_0 ($\text{m} \cdot \text{s}^{-1}$)	Multiple	2.4 ± 0.3		ES: 0.00; r : 0.97
			2-point	2.4 ± 0.2		
		P_0 (W)	Multiple	807 ± 124		ES: 0.03; r : 0.99
			2-point	803 ± 128		
Dobrijevic et al. (12)	13 collegiate men performed 2 tests consisting of walking or running on a motorized treadmill at 8 velocities ranging from 1.4 to $3.3 \text{ m} \cdot \text{s}^{-1}$.	F_0 (N)	Multiple	385 ± 88	CV: 6.4%; ICC: 0.97	ES: -0.03 ; r : 0.99
			2-point	388 ± 88	No data	
		V_0 ($\text{m} \cdot \text{s}^{-1}$)	Multiple	8.7 ± 1.6	CV: 11.1%; ICC: 0.93	ES: 0.00; r : 0.89
			2-point	8.7 ± 2.3	No data	
		a ($\text{N} \cdot \text{s} \cdot \text{m}^{-1}$)	Multiple	47.0 ± 18.5	CV: 15.7%; ICC: 0.95	ES: -0.11 ; r : 0.96
			2-point	49.2 ± 21.9	No data	
		P_0 (W)	Multiple	815 ± 140	CV: 10.0%; ICC: 0.93	ES: 0.01; r : 0.93
			2-point	814 ± 171	No data	

**Table 1
(continued)**

	15 collegiate women performed 2 tests consisting of walking or running on a motorized treadmill at 8 velocities ranging from 1.4 to 3.3 m·s ⁻¹ .	F ₀ (N)	Multiple	234 ± 53	CV: 10.9%; ICC: 0.94	ES: -0.05; r: 0.98
			2-point	237 ± 57	No data	
		V ₀ (m·s ⁻¹)	Multiple	7.3 ± 1.4	CV: 12.3%; ICC: 0.88	ES: 0.00; r: 0.93
			2-point	7.3 ± 1.6	No data	
		a (N·s·m ⁻¹)	Multiple	34.2 ± 13.2	CV: 19.3%; ICC: 0.94	ES: -0.09; r: 0.97
			2-point	35.5 ± 15.4	No data	
		P ₀ (W)	Multiple	415 ± 80	CV: 14.0%; ICC: 0.84	ES: 0.01; r: 0.93
			2-point	414 ± 74	No data	
Grbic et al. (25)	13 female students performed a test in the leg extension isokinetic exercise against 30, 60, 120, 180, and 240°·s ⁻¹ . T	F ₀ (N)	Multiple	529 ± 82	No data	ES: 0.07; r: 0.95
			2-point	522 ± 105		
		V ₀ (m·s ⁻¹)	Multiple	2.4 ± 0.4		ES: 0.17; r: 0.90
			2-point	2.3 ± 0.5		
		a (N·s·m ⁻¹)	Multiple	228 ± 37		ES: 0.12; r: 0.90
			2-point	223 ± 45		
		P ₀ (W)	Multiple	319 ± 78		ES: 0.10; r: 0.97
			2-point	308 ± 99		

Although the multiple-point method considered all the loads or velocities applied, the 2-point method used only the 2 most distant loads or velocities.

1RM = 1 repetition maximum; CMJ = countermovement jump; CV = coefficient of variation; ES = effect size (multiple-point - 2-point/[SDboth]); ICC = intraclass correlation coefficient; r = Pearson's correlation coefficient.

reduce the precision of the F-V assessment. This statement is supported by García-Ramos et al. (19) who revealed a gradual decrease in the reliability of the velocity output associated with the increment of the resistance in the leg cycle ergometer exercise. Namely, the reliability of the highest resistance was markedly lower than the one observed for the remaining resistances tested. As a consequence, when the highest resistance was considered for the construction of the 2-point method, it revealed lower reliability of the F-V relationship parameters compared with using the second highest resistance.

Although the implications of the findings presented in this section regarding the selection of the farthest data points and proximity of the points to the axis intercepts may generally stand, the optimum selection of the data points (i.e., the 2 loads or 2 velocities) should be specifically assessed for each task because the individual reliability of the 2 experimental points may markedly compromise the precision of the F-V relationship. Although no previous study has examined the optimal selection of the 2 points for modeling the L-V relationship to predict the 1RM, we recommend using 2 distant loads representing approximately 40–50% and 70–80% of self-reported 1RM. The use of 2 close loads should be avoided because it may result in a regression equation with large error. It is also important to note that although the precision in the prediction of the 1RM could be improved when the heavy load is close to the 1RM, the proposed advantages of using submaximal loads to predict the 1RM could be compromised if the loads near the maximum ones are required for the prediction (4). Future studies are needed to confirm the suitability of these recommendations to predict the 1RM.

TESTING PROCEDURES

Several methodological points should be considered for the routine testing of the F-V and L-V relationships through the 2-point method:

Warm-up. The warm-up procedure should be demanding enough to

prepare the subjects to develop their fullest potential with the 2 loads (or 2 velocities) required for the application of the 2-point method and to minimize the risk of injury. To potentiate the subjects, the warm-up should include repetitions with loads similar or even higher than the heavier load used in the F-V and L-V tests.

Sequence of the loads. Presumably, if an appropriate warm-up is conducted, the order of application of the 2 loads should not considerably influence the outcomes of the F-V and L-V relationships. This opinion is supported by different studies that have accurately determined the F-V relationship with a randomized order of the loads (20,32,53,57). Although an incremental order of the loads has been typically used to assess the L-V relationship (18,24), we believe that this is not necessary when implementing the 2-point method to predict the 1RM.

Number of trials. Two trials should be recommended for each load or velocity condition. However, when more than a 10% difference is obtained between both trials, an extra repetition should be performed and the most extreme value discarded (22). According to the most commonly applied procedures, the trial with the highest velocity or force with each load or velocity, respectively, should be used for the modeling of the relationships (25,32,43). However, further studies need to be conducted to further evaluate this problem.

Rest between trials. Adequate rest intervals are required to minimize fatigue and maximize the performance. The rest between trials may differ across the tasks. For instance, while 15–45 seconds could be enough in the bench press exercise (16,57), longer rest periods could be needed in more fatiguing tasks such as the leg cycle ergometer exercise (20,57). In addition, longer rest intervals could also be needed when lifting heavy loads compared with light loads (24). Rather than focusing on implementing fixed rest periods, strength

and conditioning professionals must ensure that the subjects are in optimal condition to perform the tasks to their fullest potential in all repetitions.

Type of variable. The averaged and maximum values of force and velocity have been used to determine the F-V and L-V relationships (13,18). Regarding the use of the L-V relationship to predict the 1RM, we recommend the use of the averaged values because the average velocity of the 1RM trial has proven to be relatively stable for a given exercise (24,35,40,50). The variability of the maximum velocity at the 1RM remains virtually unexplored. The stability of the velocity of the 1RM allows estimation of the 1RM value from the 2-point method (see next section for further details). However, less conclusive data exist regarding the preferred variable to determine the F-V relationship. García-Ramos et al. (13) reported significantly lower linearity and reliability of the F-V relationship when obtained from the averaged than from the maximum values of force and velocity in the squat jump, but no differences were observed in the countermovement jump. The assessment of the averaged values could also be compromised when obtained by linear encoders during ballistic resistance training exercises because they cannot accurately determine the end of the concentric phase (29). In this regard, the optimal variable to determine the F-V relationship may depend on both the tested task and the measurement tool used to collect the force and velocity data. Therefore, more research remains to be conducted to clarify this issue. So far, we can only generally recommend the use of the variable type that maximizes the relationship linearity because it should also maximize the reliability of the F-V parameters (13).

ANALYSIS AND INTERPRETATION OF THE RESULTS

Two straightforward examples regarding how the F-V and L-V relationships can be quickly analyzed within an Excel spreadsheet are provided in this section. The spreadsheets would

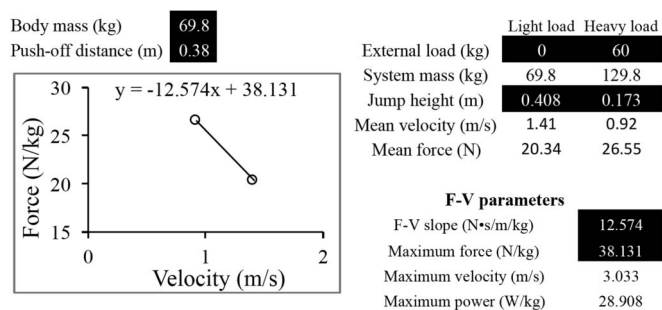


Figure 2. Image of the custom Excel spreadsheet used to determine the F-V relationship parameters through the application of the 2-point method. The Excel cells in black are the ones that should be filled by the evaluator, whereas the remaining cells are automatically completed according to the equations provided by Samozino et al. (47) (mean velocity and mean force) and the equations derived from the F-V relationship (maximum velocity = $38.131/12.574$; maximum power = $38.131 \times 3.033/4$).

provide prompt feedback information to strength and conditioning professionals that can be directly used to design and monitor the resistance training.

F-V relationship. The equations proposed by Samozino et al. (47) were used to estimate the mean values of force and velocity under 2 loading conditions: (a) unloaded squat jump and (b) the squat jump overloaded by 60 kg. Samozino's equations (47) require only 3 simple input variables: system mass (i.e., the sum of body mass and external load in kg), jump height (m), and push-off distance (m). In the example provided in Figure 2, jump height was determined from a force platform recording of the flight time, but it should be noted that the flight time can also be accurately measured by

means of more economical and practical devices (1,32).

An image of the Excel spreadsheet customized to determine the F-V relationship parameters is presented in Figure 2. The system mass is calculated as the sum of body mass and external load. Mean velocity and mean force are calculated using the equations proposed by Samozino et al. (47). Finally, the maximum velocity and maximum power capacities are calculated as $V_0 = F_0/a$ and $P_0 = F_0 \cdot V_0/4$, respectively. It should be noted that the same spreadsheet can be used when the force and velocity data are directly obtained by other measurement tools, such as force platforms, linear encoders, cycle ergometers, or isokinetic devices. In that case, the evaluator would need to indicate the force and velocity data in their respective cells, whereas the remaining

cells (i.e., body mass, push-off distance, external load, system mass, and jump height) could be deleted.

L-V relationship. Figure 3 depicts the Excel spreadsheet customized to predict the 1RM through the modeling of the individual L-V relationship obtained from the velocity recorded against 2 different external loads (i.e., the 2-point method). The first step is to put in both the magnitude of the 2 external loads and the mean velocity reached against them. Afterward, the evaluator should also indicate the slope and the load axis intercept of the line drawn through the 2 points. The velocity of the 1RM is set at $0.17 \text{ m} \cdot \text{s}^{-1}$ based on previous studies conducted on the bench press exercise (24,35,40,50). However, note that the velocity of the 1RM is exercise-dependent (7,50), and therefore, this value should be modified accordingly to the exercise evaluated.

An almost perfect concurrent validity (trivial effect sizes [from 0.02 to 0.17] and very high correlations [r ranged from 0.96 to 0.98]) of the bench press 1RM predicted by the 2-point method has been reported with respect to the directly measured 1RM (14). Note also that the validity of the 2-point method here was higher than the 1RM prediction from the generalized group equations proposed by González-Badillo and Sánchez-Medina (24). These results specifically support the use of the 2-point method to predict the 1RM. However, future studies should examine whether the 2-point method can also accurately predict the 1RM in other basic resistance training exercises (e.g., squat, leg press, bench pull, pull-up, shoulder military press, etc.).

MEASUREMENT TOOLS

A variety of devices such as force platforms (9,13,33), linear position transducers (15,44), contact platforms (22,32), motorized treadmills (12,30), isokinetic equipment (25), cycle ergometers (20,54), accelerometers (21,45), or mobile applications (1,3) have been used to determine the F-V relationship from a variety of functional tasks. The

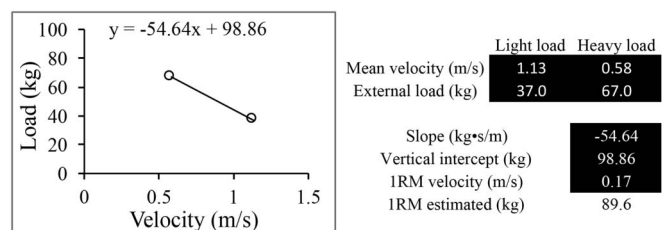


Figure 3. Image of the custom Excel spreadsheet used to determine the 1RM through the application of the 2-point method. The Excel cells in black are the ones that should be filled by the evaluator, whereas the remaining cells are automatically completed according to the following equation: $1\text{RM estimated (kg)} = -54.64 \times 0.17 + 98.86$.

vertical jump is commonly used to explore the F-V relationship of lower-body muscles (9,13,32,33,46,48). Although force platforms are considered the gold-standard for testing the F-V relationship during vertical jumps (9,21,29), their use is restricted to laboratory conditions. More practical and cost-effective devices (e.g., linear position transducers, accelerometers, and mobile apps) or simple field methods (45,47) can be routinely used by strength and conditioning professionals to assess the F-V relationship at a training venue. For instance, based on Samozino's equations (47) and the proposed optimal F-V profile (48), "My Jump 2" is a recently validated mobile application that can be used to assess the F-V relationship during vertical jumps (1).

Linear position transducers are the gold-standard for assessing L-V relationships during resistance training exercises (26). Although they are less expensive than force platforms, their price can also limit their use by strength and conditioning professionals. Although the accuracy of accelerometers to measure movement velocity has been questioned (5), the high speed video cameras of the current mobile phones can be used for that purpose. For instance, a mobile application named "PowerLift" has been recently validated to record the velocity of the barbell in the bench press exercise and, consequently, to predict the 1RM from the individual L-V relationship (3).

CONCLUSIONS AND PRACTICAL APPLICATIONS

It should be acknowledged that the 2-point method is a longer procedure than standard tests typically conducted under a single mechanical condition. However, the addition of just 1 more mechanical condition to the standard test provides outcomes of higher informative value. Namely, although the standard test is not able to determine the distinctive capacities of the muscles to produce force, velocity, and power, the 2-point method allows to assess these capacities through the

modeling of the force and velocity data collected under 2 different mechanical conditions. Although generalized group equations have been proposed to estimate the 1RM from the velocity recorded against a single loading condition, these equations provide a less accurate estimation of the 1RM than the 2-point method because they do not take into account the between-subjects differences in the load-velocity profile. Therefore, the 2-point method represents a worthy approach for both the routine testing of muscle mechanical capacities through the F-V relationship and for predicting the 1RM through the L-V relationship. The use of the 2-point method is justified by the high linearity of the F-V and L-V relationships. A number of specific recommendations that may help strength and conditioning professionals to implement the 2-point method are provided below.

FORCE-VELOCITY RELATIONSHIP

It is recommended that the 2-point method is based on 2 distant points, where 1 point should be located close to the force intercept and another 1 close to the velocity intercept. However, the extreme points may need to be avoided in some tests because of their lower reliability. A low reliability of one of the experimental points may compromise the accuracy of the 2-point method. The sequence of application of the 2 loads/velocities should not meaningfully influence the outcomes of the F-V relationship if an appropriate warm-up procedure is conducted. Two to three trials should be performed with each load providing time for full recovery between them. Finally, the appropriateness of the selection of maximum or average values for the F-V modeling should be assessed for each task. We recommend the use of the type of variable that maximizes the linearity of the F-V relationship.

LOAD-VELOCITY RELATIONSHIP

The 2-point method could also be a quick and valid procedure to predict the 1RM in different resistance

training exercises. The recommendations provided for the assessment of the F-V relationship regarding the warm-up, sequence of the loads, number of trials, and rest between trials are also applicable to the L-V relationship. The available data encourage the use of averaged over the maximum velocity values because of higher linearity of the L-V relationship and their more stable velocity of the 1RM for a given exercise. The application of the 2-point method to predict the 1RM requires 3 simple steps: (a) to set the task specific velocity of the 1RM preferably recorded in previous studies, (b) record the mean velocity against 2 different external loads representing approximately 50 and 80% of self-reported 1RM, and (c) model the individual load-velocity relationship through these 2 points and determine the 1RM as the load associated with the already set velocity of the 1RM.

Conflicts of Interest and Source of Funding: The authors report no conflicts of interest and no source of funding.



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